AEGIS Digital Umbrella - Feature Guide

Public website and appearance

- Landing page explaining the application and its scan workflow.
- Shared navigation, About, and Contact pages.
- Light and dark theme control shared across pages.
- Responsive layouts for desktop, tablet, and mobile screens.

Accounts and profile

- Account registration and sign-in.
- Forgot-password request and time-limited email reset flow when SMTP is configured.
- Profile viewing and editing, plus password change.
- Role-aware routes and administrative screens for configured administrator accounts.

Dashboard

- Summary of total scans, active scans, detected vulnerabilities, and high-risk findings.
- Recent scans with target URL, date, status, and severity totals.
- Search by target URL and filter scans by status.
- Shortcuts to start a scan and open the AI assistant.
- Scan-type overview and service status information.

Built-in website checks

The scanner checks three categories in a bounded crawl of the authorized target:

- SQL injection indicators from safe test inputs and database error responses.
- Reflected cross-site scripting indicators when a harmless probe is reflected without HTML encoding.
- CSRF token presence on state-changing forms, based on recognizable token field names.

The current scanner is a built-in, limited checker. It is not OWASP ZAP or Burp Suite, and it does not provide the same depth of crawling or verification. Findings are indicators for review; custom defenses may be missed and false positives are possible.

Results and remediation

- Scan summary with total findings and High, Medium, and Low counts.

• Coverage information including visited pages and tested input fields.

• Per-finding category, severity, location, evidence, and description.

• Gemini-generated remediation guidance when a valid API key and provider response are available.

• Clear status when AI guidance is unavailable; findings remain visible independently.

• Downloadable PDF reports for individual scans.

Scan history and reports

• Saved scan history associated with the signed-in user.

• Search and status filters for reviewing scans.

• CSV and PDF history export, plus a PDF report for an individual scan.

AI assistant

• Cybersecurity and IT assistant focused on defensive security, OWASP guidance, vulnerability assessment, authorized testing, and secure coding.

• Requires a valid Gemini API key in the backend environment. The assistant may be unavailable if the provider rejects the key or is temporarily unavailable.

Administration

• Administrative views for user accounts, scans, and summary statistics.

• Access depends on the account role and backend authorization configuration.

Service requirements

Capability | Requirement

Account and scan storage | MongoDB connection in `backend/.env`

AI assistant and recommendations | `GEMINI_API_KEY` and Gemini service access

Password reset email | Valid SMTP settings; Gmail users should use an App Password

Frontend and scanner | Node.js/npm, Python dependencies, and the backend running locally

Authorized use

Only scan websites you own or have explicit authorization to assess. The scanner sends test requests. A clean scan is not proof that a target is secure; interpret results as signals and verify them through an authorized security review.